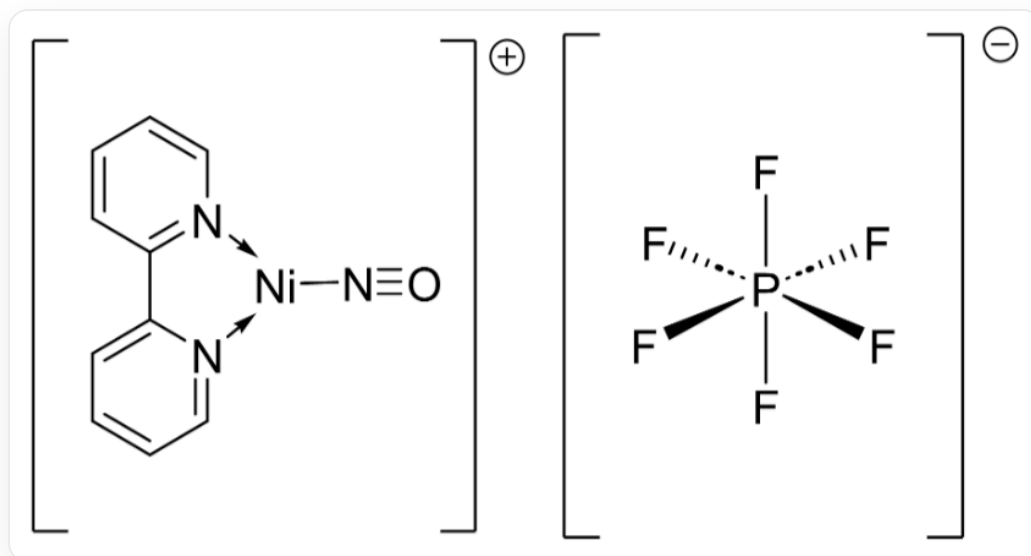


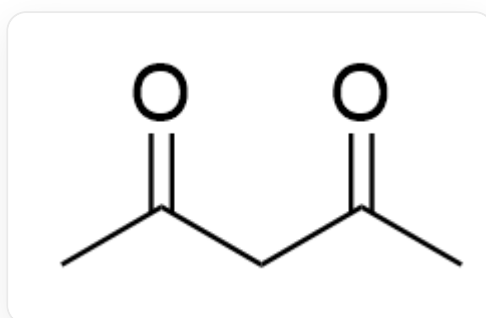
Question

Researchers synthesized and analyzed the chemical reaction of complex **A**, the structure of complex **A** is shown in the following figure:



This complex is a 1:1 zwitterionic salt with each ion carrying a single charge. The central metal of the cation is Ni with planar triangular coordination, where two coordination sites are coordinated by 2,2'-bipyridine, and one site is coordinated by N of linear NO; the anion has P as the center, coordinated by 6 F in an octahedral arrangement

Complex **A** can further coordinate with a neutral ligand in another **A** to obtain complex **B**. The following formula ligand **X** is added to the latter for reaction:



SMILES: CC(CC(C)=O)=O

At this time, a chemical reaction occurs to obtain a gas with a density of 1.97 g/L under standard conditions.

It was found that this is a multi-step reaction: two molecules of complex **B** first undergo a ligand substitution reaction to obtain complexes **C** and **D**, and **D** contains a N = N double bond; then, complex **D** undergoes a ligand decomposition reaction, and further reacts with the newly added ligand **X** and with complex **C**, finally obtaining a transition metal-containing complex **E**. The above notations **A** ~ **E** are all complete chemical formulas of the compounds.

The following propositions are given:

1. The total number of electrons around the transition metal in complex **A** is the same as that in complex **D**.
2. The symmetry of the complex in **D** is higher than that of complex **B**.
3. In complex **B**, considering the most likely situation, the valence state of the transition metal is 0.
4. In the chemical formula of one complex **C**, the sum of the point group orders of all ionic species is 102.
5. Among the complexes **A** ~ **E**, only one is electrically neutral.
6. For every 22.4 L of gas at standard temperature and pressure produced in the reaction, approximately 18 mL of small molecule by-products are obtained at room temperature and normal pressure.

Please calculate the ratio of the sum of the serial numbers of the correct propositions to (the sum of the serial numbers of the incorrect propositions + 1), and choose the correct option.

A. 0.375

B. 2.143

C. 0.467

D. 1

E. 1.2

F. 0.692

G. 4.5

H. 0

I. 21

Answer

Correct Answer: **G**

Detailed Explanation

First, based on the information given in the question, deduce the chemical formulas and properties of each species.

Complex **A**:

- Cation: $[\text{Ni}(\text{bpy})(\text{NO})]^+$. Where, bpy (2,2'-bipyridine) is a neutral ligand. The NO ligand is described as linear, which usually means it coordinates as NO^+ (nitrosonium ion), a 2-electron donor. To make the cation have a +1 charge, the oxidation state of Ni must be 0. That is, $[\text{Ni}^0(\text{bpy})(\text{NO}^+)]^+$.

- Anion: $[\text{PF}_6]^-$.

- Chemical formula: $[\text{Ni}(\text{bpy})(\text{NO})][\text{PF}_6]$.

CHECKPOINT

1 PTS

Complex **A** is $[\text{Ni}(\text{bpy})(\text{NO})][\text{PF}_6]$

Complex **B**:

- Obtained from the reaction of complex **A** with another neutral ligand bpy:



- The chemical formula of complex **B** is $[\text{Ni}(\text{bpy})_2(\text{NO})][\text{PF}_6]$.

CHECKPOINT

1 PTS

Complex **B** is $[\text{Ni}(\text{bpy})_2(\text{NO})][\text{PF}_6]$

Gaseous product of the reaction:

- The reaction produces a gas with a density of 1.97 g/L under standard conditions (STP, 0°C, 1 atm).
- The molar mass of the gas $M = \text{density} \times V_m = 1.97 \text{ g/L} \times 22.4 \text{ L/mol} \approx 44.13 \text{ g/mol}$.
- This value is very close to the molar mass of nitrous oxide (N_2O) or carbon dioxide (CO_2) (both are 44.02 g/mol). Combined with the information of ligand NO and the presence of $\text{N}=\text{N}$ double bond in the complex in the question. Therefore, the gas produced is N_2O .

CHECKPOINT

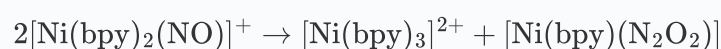
1 PTS

The gaseous product of the reaction is N_2O

Multi-step reaction mechanism:

First step: $2\text{B} \rightarrow \text{C} + \text{D}$

- This is a ligand redistribution reaction starting from two molecules of cation $[\text{Ni}(\text{bpy})_2(\text{NO})]^+$.
- Complex **D** contains $\text{N}=\text{N}$ double bond, indicating that two NO ligands in **D** have undergone coupling, the reaction is:



From this, **C** and **D** can be determined:

- **C**: Ionic complex $[\text{Ni}(\text{bpy})_3][\text{PF}_6]_2$

CHECKPOINT

1 PTS

Complex **C** is $[\text{Ni}(\text{bpy})_3][\text{PF}_6]_2$

- **D**: Neutral complex $[\text{Ni}(\text{bpy})(\text{N}_2\text{O}_2)]$

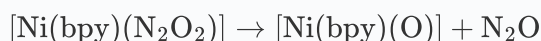
CHECKPOINT

1 PTS

Complex **D** is $[\text{Ni}(\text{bpy})(\text{N}_2\text{O}_2)]$

Second step: **D** decomposes and reacts with **X**

- First, complex **D** decomposes. The N_2O_2 ligand therein undergoes a decomposition reaction to generate gas N_2O and an oxo species:



- The ligand **X** (i.e., acacH , acetylacetone) does not directly react with the above-generated oxo species $[\text{Ni}(\text{bpy})(\text{O})]$, but is related to the final product. In fact, there are several possible pathways in the reaction system. A more reasonable mechanism is that $[\text{Ni}(\text{bpy})(\text{O})]$ reacts with complex **C**, and ligand **X** participates in the subsequent steps. To simplify and obtain the final product, we consider the overall reaction:



- Therefore, the final product **E** is $[\text{Ni}(\text{bpy})_2(\text{acac})][\text{PF}_6]$.

CHECKPOINT

1 PTS

Complex **E** is $[\text{Ni}(\text{bpy})_2(\text{acac})][\text{PF}_6]$

Proposition Judgment

1. The total number of electrons around the transition metal in complex **A** is the same as that in complex **D**

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- **A** (cation): $[\text{Ni}^0(\text{bpy})(\text{NO}^+)]^+$. Total number of electrons = $\text{Ni}^0 (d^{10}, 10e^-) + \text{bpy} (4e^-) + \text{NO}^+ (2e^-) = 16e^-$.

- **D**: $[\text{Ni}(\text{bpy})(\text{N}_2\text{O}_2)]$. To maintain electroneutrality, if the ligand is $\text{N}_2\text{O}_2^{2-}$, then the metal is Ni^{2+} . Total number of electrons = $\text{Ni}^{2+} (d^8, 8e^-) + \text{bpy} (4e^-) + \text{N}_2\text{O}_2^{2-} (4e^-) = 16e^-$.

- Conclusion: Proposition 1 is correct.

CHECKPOINT

1 PTS

The total number of electrons around the transition metal in complex **A** and complex **D** are both 16-electron structure

2. The symmetry of the complex in **D** is higher than that of the complex in **B**.

- First, analyze the valence state of the metal in the cation in complex **B**: $[\text{Ni}(\text{bpy})_2(\text{NO})]^+$. If the metal is in the 0 valence state, then it is Ni^0 (d^{10} , $10e^-$), and the ligand is NO^+ , the total number of electrons is $10e^- + 2 \times \text{bpy} (8e^-) + \text{NO}^+ (2e^-) = 20e^-$. This does not conform to the common 18-electron rule. If the metal is in the +2 valence state, then it is Ni^{2+} (d^8 , $8e^-$), and the ligand is NO^- (bent, 2e- donor), the total number of electrons is $8e^- + 8e^- + 2e^- = 18e^-$. The 18-electron structure is more stable, so the metal valence state is more likely to be +2 valence.
- Complex **D** is $[\text{Ni}(\text{bpy})(\text{N}_2\text{O}_2)]$, in which the N_2O_2 ligand forms a five-membered ring with the metal, and the arrangement of the four coordination sites is planar quadrilateral, and the whole molecule has C_{2v} symmetry.
- Complex **B** has a bent coordinated NO^- in the cation. Considering the possibility that its five coordination sites are arranged in a square pyramidal or trigonal bipyramidal arrangement, it is impossible to achieve a complex cation structure with a symmetry higher than or equal to C_{2v} .
- Conclusion: Proposition 2 is correct.

CHECKPOINT

1 PTS

The point group of the complex in **D** is C_{2v} , which is higher than complex **B**

3. In complex **B**, considering the most likely situation, the valence state of the transition metal is 0 valence.

- According to the previous analysis, the metal is Ni^{2+} (d^8 , $8e^-$) with a valence of +2, and the total number of electrons is $8e^- + 8e^- + 2e^- = 18e^-$. The 18-electron structure is more stable, so the metal valence state is more likely to be +2 valence.
- Conclusion: Proposition 3 is incorrect.

CHECKPOINT

1 PTS

In complex **B**, considering the most likely situation, the valence state of the transition metal is +2 valence

4. In the chemical formula of one complex **C, the sum of the orders of the point groups of all ionic species is 102.**

- The chemical formula of complex **C** is $[\text{Ni}(\text{bpy})_3][\text{PF}_6]_2$, where the cation $[\text{Ni}(\text{bpy})_3]^{2+}$ has D_3 symmetry (group of order 6) and the anion $[\text{PF}_6]^-$ has O_h symmetry (group of order 48).
- Since there are two anions in the chemical formula, the sum of the orders of the point groups of all ionic species $= 6 + (48 \times 2) = 102$.
- Conclusion: Proposition 4 is correct.

CHECKPOINT

1 PTS

In the chemical formula of one complex **C**, the sum of the orders of the point groups of all ionic species is 102

5. Among complexes **A ~ **E**, only one is neutral.**

- **A**, **B**, **C**, **E** are all ionic salts containing $[\text{PF}_6]^-$ counterions. Complex **D**, i.e., $[\text{Ni}(\text{bpy})(\text{N}_2\text{O}_2)]$, is an uncharged inner complex.
- Conclusion: Proposition 5 is correct.

CHECKPOINT

1 PTS

Among complexes **A** ~ **E**, only **D** is a neutral complex

6. For every 22.4 L of gas at standard temperature and pressure produced in the reaction, approximately 18 mL of small molecule by-product is obtained under ambient conditions.

- According to the mechanism derivation, for every 1 mol of gas (N_2O) produced, 2 mol of acacH will be consumed in the subsequent steps and 1 mol of small molecule by-product (H_2O) will be generated.
- 1 mol of N_2O has a volume of 22.4 L at STP.
- 1 mol of H_2O has a mass of 18 g, which is liquid under ambient conditions with a density of approximately 1 g/mL and a volume of approximately 18 mL.
- Conclusion: Proposition 6 is correct.

CHECKPOINT

1 PTS

For every 22.4 L of N_2O at standard temperature and pressure produced in the reaction, approximately 18 mL of H_2O is obtained under ambient conditions

Calculate the final result:

- Correct proposition numbers: 1, 2, 4, 5, 6. The sum is $1 + 2 + 4 + 5 + 6 = 18$.
- Incorrect proposition number: 3. The sum is 3.
- Calculate the ratio:

$$\frac{\text{Sum of correct proposition numbers}}{\text{Sum of incorrect proposition numbers} + 1} = \frac{18}{3 + 1} = \frac{18}{4} = 4.5$$

Therefore, choose option **G**.

CHECKPOINT

1 PTS

The final calculated result is 4.5

